A Formal Analysis of the SENA Fusion Matrix: A Typed Supra-Adjacency Model for Social-Epistemic Network Analysis

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Abstract

Social Network Analysis (SNA) represents relations among persons, while Epistemic Network Analysis (ENA) represents relations among coded epistemic elements. The proposed SENA fusion formula combines a person-person social matrix, a code-code epistemic co-occurrence matrix, and a person-code bridge matrix into one block matrix. This paper gives a formal mathematical analysis of that formula. We show that, under explicit dimensional, nonnegativity, symmetry, and normalization assumptions, the SENA fusion matrix is a valid weighted typed adjacency matrix on a heterogeneous social-epistemic graph. We prove dimensional consistency, symmetry conditions, Laplacian positive semidefiniteness, coherent boundary cases, and the distinction between a joint adjacency model and an ENA projection. We also identify failure modes and statistical validation requirements. The main conclusion is that SENA should be interpreted not as a simple visual overlay of SNA and ENA, but as a typed supra-adjacency representation of a social-epistemic nexus. Its novelty is domain-specific formalization and validation, not the invention of block adjacency matrices themselves.

Keywords: social network analysis; epistemic network analysis; heterogeneous networks; multilayer networks; supra-adjacency matrix; collaborative learning analytics; SENA.

Mathematics Subject Classification: 05C50, 05C82, 62H25, 91D30.

Introduction

Collaborative learning is simultaneously social and epistemic. Learners interact with particular people while producing, connecting, challenging, and refining particular ideas. SNA gives mature tools for person-person relations, such as degree, density, centrality, communities, and brokerage . ENA gives a formal way to model co-occurrence relations among coded epistemic elements and to project units of analysis into a low-dimensional epistemic space .

The methodological gap is not that SNA and ENA cannot both be applied to the same data. Prior work such as SENS and iSENS already argues for combining social and epistemic perspectives . The sharper mathematical question is whether one can construct a single object that contains person-person, code-code, and person-code relations while preserving their type distinctions and while remaining compatible with standard network analysis.

The proposed SENA formula is

$$A_{fusion} = \begin{bmatrix} \alpha \hat{S} & \gamma \hat{B} \\ \gamma \hat{B}^{\top} & \beta \widehat{W} \end{bmatrix},$$

where $\hat{S}$ is a normalized social matrix, $\widehat{W}$ is a normalized epistemic co-occurrence matrix, $\hat{B}$ is a normalized person-code bridge matrix, and $\alpha, \beta, \gamma \geq 0$ are layer weights.

This paper analyzes the fusion matrix as a mathematical object. The central claim is:

> The SENA formula is mathematically valid as a normalized weighted heterogeneous adjacency matrix on the typed node set $P \cup C$. It is not, by itself, an ENA projection, a causal model, or an inferential test.

Related Mathematical and Methodological Context

The formula sits at the intersection of four established literatures. First, SNA supplies the person-person graph layer . Second, ENA supplies the code-code co-occurrence logic and, separately, projection methods for units of analysis . Third, multilayer network theory supplies supra-adjacency representations for systems with multiple layers or relation types . Fourth, heterogeneous information network theory supplies typed nodes and typed edges .

SENS and iSENS are the nearest learning-analytics precedents. They establish that SNA and ENA can provide complementary information about collaborative learning roles, discourse, group differences, and performance . However, their main contribution is an integrated analytic strategy rather than the specific block object represented by the SENA fusion matrix. Conversely, multilayer and heterogeneous-network theory make this block construction mathematically unsurprising, but they do not supply the learning-analytics semantics of S, W, B, and G.

Consequently, a calibrated novelty claim is required. SENA does not invent block matrices or multilayer graphs. Its contribution is the domain-specific formalization of a social-epistemic graph that keeps persons and epistemic codes as different node types while retaining social ties, epistemic co-occurrences, and contribution bridges in one traceable model object.

Notation and Empirical Matrices

Let

$$P = \{p_1, \ldots, p_n\}, C = \{c_1, \ldots, c_m\}$$

denote the person set and the code set. The typed node set is

$$V = P \cup C, |V| = n + m, P \cap C = \varnothing.$$

Definition 1 (Social layer). The social layer is a matrix

$$S \in \mathbb{R}_{\geq 0}^{n \times n},$$

where S_{ij} denotes the observed interaction strength from p_i to p_j. In an undirected model, $S_{ij} = S_{ji}$.

Definition 2 (Epistemic co-occurrence layer). The epistemic layer is a matrix

$$W \in \mathbb{R}_{\geq 0}^{m \times m},$$

where W_{ab} denotes the co-occurrence strength between codes c_a and c_b within a specified unit, stanza, or moving window. In the usual ENA co-occurrence construction, $W_{ab} = W_{ba}$ and diagonal entries are either zero or treated separately as code frequencies.

Definition 3 (Bridge layer). The bridge layer is a matrix

$$B \in \mathbb{R}_{\geq 0}^{n \times m},$$

where B_{ia} denotes the contribution, use, or activation strength of code c_a by person p_i.

Definition 4 (Layer normalization). A layer normalization is a map $N_L : M_L \mapsto \widehat{M}_L$ applied separately to $L \in \{S, W, B\}$. Examples include

$$\widehat{M} = \frac{M}{\max_{uv} |M_{uv}|}, \widehat{M} = \frac{\log(1+M)}{\max_{uv} \log(1+M_{uv})}, \widehat{M} = \frac{M}{\|M\|_F}.$$

A Generative Construction

Let $T = \{t_1, \dots, t_q\}$ be a set of discourse windows. Let

$$X_{ta} = \text{activation strength of code } c_a \text{ in window } t,$$

$$Y_{it} = \text{participation strength of person } p_i \text{ in window } t,$$

and let R_{ij} be the observed social interaction strength from p_i to p_j. A common SENA-compatible construction is

$$S_{ij} = R_{ij} \text{ or } S_{ij} = R_{ij} + R_{ji}.$$

$$W_{ab} = \sum_{t=1}^{q} X_{ta} X_{tb}, a \neq b.$$

$$B_{ia} = \sum_{t=1}^{q} Y_{it} X_{ta}.$$

The second equation is an ENA-style code co-occurrence object before dimensional reduction. The third equation is an affiliation or contribution object connecting persons to codes.

Assumptions

Assumption 1 (Dimensional compatibility). $S \in \mathbb{R}^{n \times n}$, $W \in \mathbb{R}^{m \times m}$, and $B \in \mathbb{R}^{n \times m}$.

Assumption 2 (Nonnegativity). S, W, B have nonnegative entries and $\alpha, \beta, \gamma \geq 0$.

Assumption 3 (Undirected baseline). For the undirected SENA baseline, $S = S^\top$ and $W = W^\top$.

Assumption 4 (Explicit normalization). The matrices entering the fusion matrix are normalized within layers before cross-layer scaling by α, β, γ.

Assumption 5 (Typed-node semantics). The labels p_i and c_a belong to different node types. Distances or centralities computed from A_{fusion} must be interpreted with respect to this typed graph, not as if all nodes were semantically identical.

Main Results

Proposition 1 (Dimensional consistency). *Under Assumption 1, the matrix* A_{fusion} *is an* $(n+m) \times (n+m)$ *matrix.*

Proof. The upper-left block has n rows and n columns. The upper-right block has n rows and m columns. The lower-left block has m rows and n columns. The lower-right block has m rows and m columns. Therefore the block matrix has $n+m$ rows and $n+m$ columns. □

Theorem 1 (Weighted undirected heterogeneous graph). *Under Assumptions 1–3,* A_{fusion} *is a symmetric nonnegative adjacency matrix on the typed node set* $V = P \cup C$.

Proof. Nonnegativity follows from nonnegative entries and nonnegative layer weights. For symmetry,

$$A_{fusion}^\top = \begin{bmatrix} (\alpha \hat{S})^\top & (\gamma \hat{B}^\top)^\top \\ (\gamma \hat{B})^\top & (\beta \widehat{W})^\top \end{bmatrix} = \begin{bmatrix} \alpha \hat{S} & \gamma \hat{B} \\ \gamma \hat{B}^\top & \beta \widehat{W} \end{bmatrix} = A_{fusion},$$

because $\hat{S} = \hat{S}^\top$ and $\widehat{W} = \widehat{W}^\top$. The block locations define three edge types: person-person, code-code, and person-code. Hence A_{fusion} is the adjacency matrix of a weighted undirected heterogeneous graph. □

Corollary 1 (Laplacian positive semidefiniteness). *Let* $D = diag(d_u)$*, where* $d_u = \sum_v A_{uv}$*, and define* $L = D - A_{fusion}$*. Under the assumptions of the theorem,* L *is positive semidefinite.*

Proof. For any $x \in \mathbb{R}^{n+m}$,

$$x^\top L x = \frac{1}{2} \sum_{u,v} A_{uv} (x_u - x_v)^2 \geq 0,$$

because $A_{uv} \geq 0$. Thus $L \succcurlyeq 0$. □

Remark 1. This result concerns the graph Laplacian, not the adjacency matrix itself. A nonnegative symmetric adjacency matrix need not be positive semidefinite. For example, $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ has eigenvalues 1 and -1.

Proposition 2 (Coherent boundary cases). *The SENA formula has the following mathematically coherent limits:*

1. *If $\gamma = 0$, the social and epistemic layers are disconnected.*
2. *If $\alpha = \beta = 0$, the graph is a pure person-code bipartite graph.*
3. *If $B = 0$, the formula reduces to a disconnected social block and epistemic block.*
4. *If $\alpha = 0$, person-person social edges are suppressed.*
5. *If $\beta = 0$, code-code epistemic edges are suppressed.*

Proof. Each statement follows by substituting the specified zero value into the fusion matrix and reading the remaining nonzero blocks. □

Theorem 2 (No direct contradiction with ENA projection). *The claim that SNA and ENA cannot be shown in one model because SNA is not dimensionally reduced while ENA is dimensionally reduced is false when SENA is constructed as the SENA fusion matrix and then embedded by a declared graph embedding operator.*

Proof. Let Φ be an embedding operator, for example spectral embedding, Laplacian eigenmaps, multidimensional scaling on graph distances, or a specified force-directed layout. The displayed coordinates are then

$$Z = \Phi\left(A_{fusion}\right) \in \mathbb{R}^{(n+m)\times d}.$$

The coordinates are derived from the joint object A_{fusion}, not by placing raw SNA nodes into a pre-existing ENA projection. Hence the dimensionality of a prior ENA display does not prevent a joint SENA embedding. What is invalid is a different operation: visually overlaying a raw social graph onto an ENA space and then interpreting all cross-type distances as if they came from one common model. □

Person-Code-Pair Attribution

The bridge matrix B answers which person contributed to which code. ENA edges, however, are code-code relations. To attribute an ENA edge to persons, define a person-code-pair tensor or matrix

$$G_{i,ab} = \text{contribution of } p_i \text{ to the code pair } \left(c_a, c_b\right).$$

A compatible estimator is

$$G_{i,ab} = \sum_{t=1}^{q} Y_{it} X_{ta} X_{tb}.$$

Thus B is appropriate for person-code bridge visualization, while G is more appropriate for claims such as “person i supported the Evidence-Explanation connection.” The current SENA implementation includes such a G block for reporting person-code-pair contribution.

Normalization and Identifiability

The layer weights α, β, γ are interpretable only relative to the normalization. If S, W, B are rescaled arbitrarily, then the same visual or graph-theoretic output can be produced by compensating changes in α, β, γ. Thus the parameters are not intrinsically identifiable without a normalization convention.

Proposition 3 (Scale dependence). *Let $c_S, c_W, c_B > 0$. Replacing S, W, B by $c_S S, c_W W, c_B B$ and replacing α, β, γ by $\alpha / c_S, \beta / c_W, \gamma / c_B$ leaves A_{fusion} unchanged.*

Proof. Substitute the rescaled matrices and rescaled weights into the fusion matrix. Each block is unchanged. □

Corollary 2. *Any empirical SENA report must disclose the normalization rule before interpreting α, β, γ.*

Directed and Temporal Extensions

If $S \neq S^{\top}$, then the same block construction defines a directed heterogeneous graph rather than an undirected one. The Laplacian result above does not apply without additional directed-graph choices or symmetrization.

A more general directed bridge model is

$$A_{directed} = \begin{bmatrix} \alpha \hat{S} & \gamma_{out} \hat{B}^{PC} \\ \gamma_{in} \hat{B}^{CP} & \beta \widehat{W} \end{bmatrix},$$

where B^{PC} represents person-to-code activation and B^{CP} represents code-to-person uptake, attribution, feedback, or recommendation.

For temporal analysis, define

$$A_{fusion}(t) = \begin{bmatrix} \alpha \hat{S}(t) & \gamma \hat{B}(t) \\ \gamma \hat{B}(t)^{\top} & \beta \widehat{W}(t) \end{bmatrix}.$$

Then temporal change can be studied using

$$\Delta A(t_1, t_2) = A_{fusion}(t_2) - A_{fusion}(t_1),$$

or by distances such as Frobenius distance, spectral distance, graph edit distance, or permutation-tested group/time contrasts.

Failure Modes and Counterexamples

Example 1 (Projection error). If a researcher places person nodes from an SNA graph onto a previously computed ENA unit space and interprets person-code distances as distances from a common model, the interpretation is invalid. The coordinates were not jointly estimated from A_{fusion}.

Example 2 (Layer dominance). If S contains counts in the hundreds while W and B contain binary values, then centrality or layout on the unnormalized matrix is dominated by the social layer. This is a scale artifact, not a substantive social-epistemic finding.

Example 3 (Coding unreliability). If code assignments are unstable, then W, B, and G inherit that instability. A formally valid matrix can still be empirically unreliable.

Statistical Validation Framework

A publication-grade SENA study should report:

1. the data contract: persons, interactions, windows/stanzas, coded segments, and codebook;
2. exact formulas for S, W, B, and, if used, G;
3. the normalization rule;
4. sensitivity analyses over (α, β, γ);
5. the embedding operator Φ, including random seed if stochastic;
6. coding reliability or human-reviewed AI coding procedures;
7. evidence traceability from each edge to source records;
8. null models or resampling tests for group and temporal claims.

Useful null models include code-label shuffling within learners, social-edge rewiring with degree preservation, stanza membership shuffling within temporal bands, group-label permutation, and bootstrap resampling of windows.

Discussion

The formal analysis supports the mathematical reasonableness of the SENA formula, but it also narrows the permissible interpretation. The formula does not magically convert SNA and ENA into one latent space. Rather, it constructs a joint typed graph. A shared coordinate system becomes meaningful only after a specified embedding or matrix factorization is applied to that joint graph.

The most defensible theoretical language is therefore:

> SENA represents collaborative learning as a typed social-epistemic graph whose supra-adjacency matrix combines person-person social ties, code-code epistemic co-occurrences, and person-code contribution bridges.

This wording connects SENA to established mathematics while preserving its substantive contribution for learning analytics.

Conclusion

Under explicit dimensional, nonnegativity, symmetry, and normalization conditions, the SENA fusion matrix

$$A_{fusion} = \begin{bmatrix} \alpha \hat{S} & \gamma \hat{B} \\ \gamma \hat{B}^{T} & \beta \widehat{W} \end{bmatrix}$$

is a valid weighted typed adjacency matrix. Its graph Laplacian is positive semidefinite in the undirected case, boundary cases reduce to recognizable SNA, ENA, and bipartite submodels, and joint embeddings are mathematically permissible when they are computed from A_{fusion} itself. The formula becomes a rigorous research method only when matrix construction, normalization, embedding, statistical testing, and evidence traceability are fully specified.

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